Product parameters

Rated voltage: 5V

Diameter of test consumables: 1.75mm

Wiring pins:



Product Features

1. It can detect abnormal extrusion of consumables caused by plugging, entanglement, extruder failure, etc.
2. It can replace ordinary material break detection, such as consumable breakage and consumable exhaustion.
3. It is suitable for open source firmware Marlin 2.0.x and Klipper
4. The installation method can be freely selected.

Instructions for use

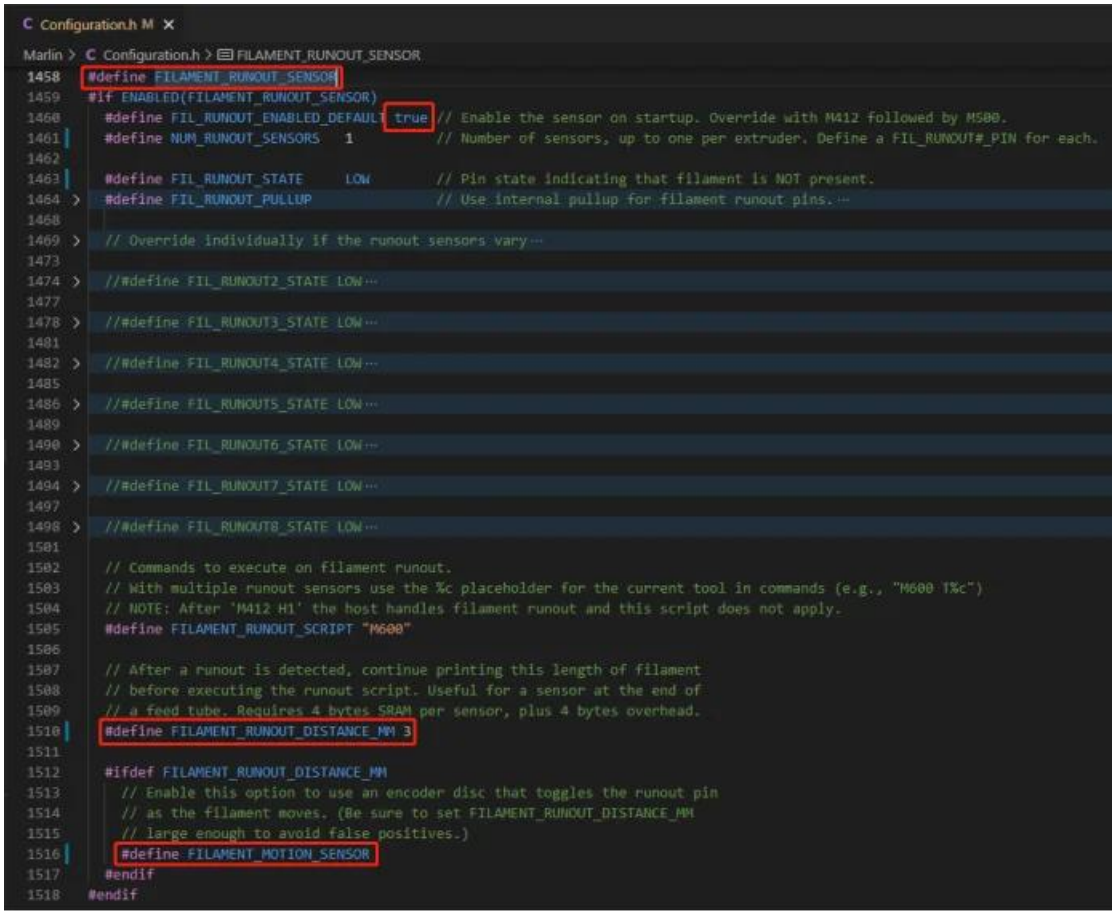
Insert the filament from the quick connector in any direction of the detection module;
When printing or discharging normally, the blue light on the module flashes. When the extruder is not working, the blue light on the module is always on or off.

Firmware Config for Filament blockage detection：

Marlin：

In configuration.h:

1. uncomment #define FILAMENT_RUNOUT_SENSOR.
2. uncomment #define FILAMENT_RUNOUT_DISTANCE_MM 7 and set it to 7mm
3. uncomment #define FILAMENT_MOTION_SENSOR



```
Configuration.h M
Marlin > Configuration.h > FILAMENT_RUNOUT_SENSOR
#define FILAMENT_RUNOUT_SENSOR
#if ENABLED(FILAMENT_RUNOUT_SENSOR)
  #define FIL_RUNOUT_ENABLED_DEFAULT true // Enable the sensor on startup. Override with M412 followed by M500.
  #define NUM_RUNOUT_SENSORS   1          // Number of sensors, up to one per extruder. Define a FIL_RUNOUT#_PIN for each.

  #define FIL_RUNOUT_STATE     LOW        // Pin state indicating that filament is NOT present.
  #define FIL_RUNOUT_PULLUP               // Use internal pullup for filament runout pins.···

  // Override individually if the runout sensors vary···

  //#define FIL_RUNOUT2_STATE LOW···

  //#define FIL_RUNOUT3_STATE LOW···

  //#define FIL_RUNOUT4_STATE LOW···

  //#define FIL_RUNOUT5_STATE LOW···

  //#define FIL_RUNOUT6_STATE LOW···

  //#define FIL_RUNOUT7_STATE LOW···

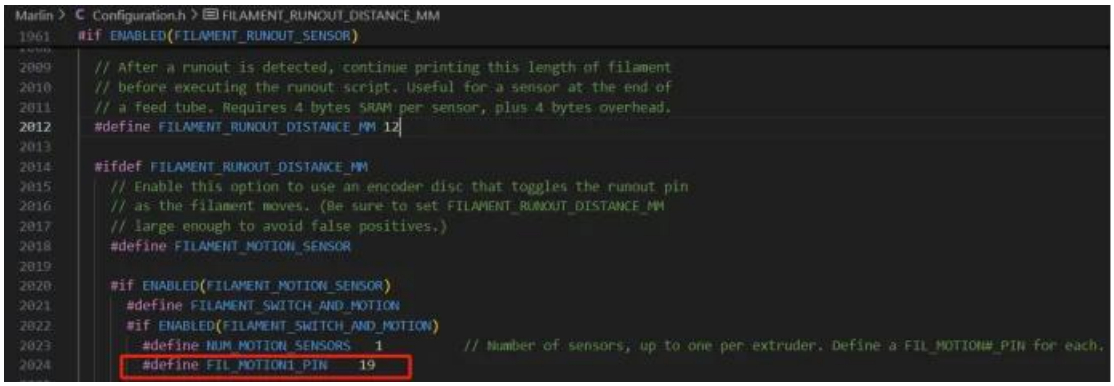
  //#define FIL_RUNOUT8_STATE LOW···

  // Commands to execute on filament runout.
  // With multiple runout sensors use the %c placeholder for the current tool in commands (e.g., "M600 T%c")
  // NOTE: After 'M412 H1' the host handles filament runout and this script does not apply.
  #define FILAMENT_RUNOUT_SCRIPT "M600"

  // After a runout is detected, continue printing this length of filament
  // before executing the runout script. Useful for a sensor at the end of
  // a feed tube. Requires 4 bytes SRAM per sensor, plus 4 bytes overhead.
  #define FILAMENT_RUNOUT_DISTANCE_MM 3

  #ifdef FILAMENT_RUNOUT_DISTANCE_MM
    // Enable this option to use an encoder disc that toggles the runout pin
    // as the filament moves. (Be sure to set FILAMENT_RUNOUT_DISTANCE_MM
    // large enough to avoid false positives.)
    #define FILAMENT_MOTION_SENSOR
  #endif
#endif
```

If your firmware The version is relatively new.Stil need set motion pin to actual GPIO of motherboard



```
Marlin > Configuration.h > FILAMENT_RUNOUT_DISTANCE_MM
#if ENABLED(FILAMENT_RUNOUT_SENSOR)

  // After a runout is detected, continue printing this length of filament
  // before executing the runout script. Useful for a sensor at the end of
  // a feed tube. Requires 4 bytes SRAM per sensor, plus 4 bytes overhead.
  #define FILAMENT_RUNOUT_DISTANCE_MM 12

  #ifdef FILAMENT_RUNOUT_DISTANCE_MM
    // Enable this option to use an encoder disc that toggles the runout pin
    // as the filament moves. (Be sure to set FILAMENT_RUNOUT_DISTANCE_MM
    // large enough to avoid false positives.)
    #define FILAMENT_MOTION_SENSOR

    #if ENABLED(FILAMENT_MOTION_SENSOR)
      #define FILAMENT_SWITCH_AND_MOTION
      #if ENABLED(FILAMENT_SWITCH_AND_MOTION)
        #define NUM_MOTION_SENSORS   1          // Number of sensors, up to one per extruder. Define a FIL_MOTION#_PIN for each.
        #define FIL_MOTION1_PIN    19
```

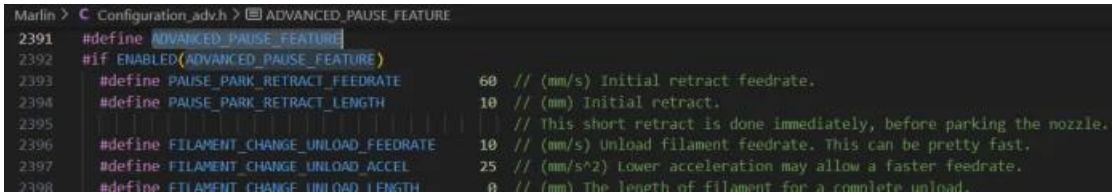
4.The filament sensor feature of Marlin also needs to enable #define NOZZLE_PARK_FEATURE

```
#define NOZZLE_PARK_FEATURE

#if ENABLED(NOZZLE_PARK_FEATURE)
  // Specify a park position as { X, Y, Z_raise }
  #define NOZZLE_PARK_POINT { (X_MIN_POS + 10), (Y_MIN_POS + 10), 0 }
  #define NOZZLE_PARK_X_ONLY          // X move only is required to park
  //#define NOZZLE_PARK_Y_ONLY        // Y move only is required to park
  #define NOZZLE_PARK_Z_RAISE_MIN   0   // (mm) Always raise Z by at least this distance
  #define NOZZLE_PARK_XY_FEEDRATE 100   // (mm/s) X and Y axes feedrate (also used for delta Z axis)
  #define NOZZLE_PARK_Z_FEEDRATE    5   // (mm/s) Z axis feedrate (not used for delta printers)
#endif
```

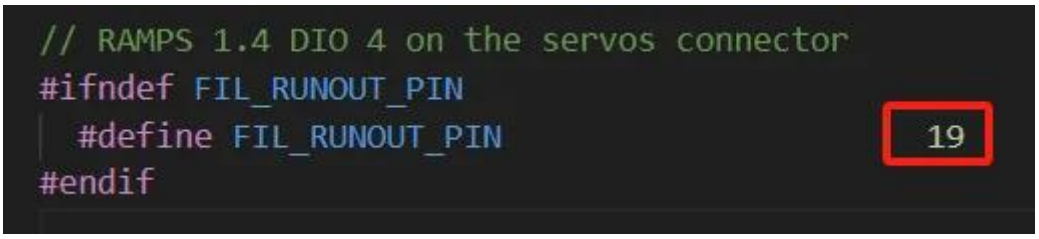
In configuration_adv.h:

uncomment #define ADVANCED_PAUSE_FEATURE

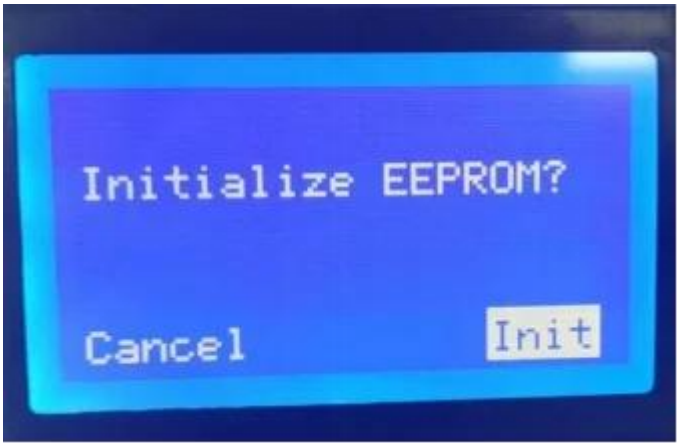


5.Check whether the FIL_RUNOUT_PINin the pin file of the used motherboard is same as theactual motion sensor connecting pin.

In Marlin/src/pin/find your pin file:



6.After uploading the firmware, it might be necessary to reinitialize the configuration in order to achieve the desired effect.



Klipper：

In printer.cfg:

Definition:

```
[filament_motion_sensor my_sensor]
detection_length: 7.0
#   The minimum length of filament pulled through the sensor to trigger
#   a state change on the switch_pin
#   Default is 7 mm.
extruder:
#   The name of the extruder section this sensor is associated with.
#   This parameter must be provided.
switch_pin:
#pause_on_runout:
#runout_gcode:
#insert_gcode:
#event_delay:
#pause_delay:
#   See the "filament_switch_sensor" section for a description of the
#   above parameters.
```

Example： Ideaformer IR3_V2

```
View 'filament_motion_sensor' documentation
[filament_motion_sensor encoder_sensor]
switch_pin: PA1
detection_length: 12
extruder: extruder
pause_on_runout: True
runout_gcode:
  BASE_PAUSE
    M117 Filament encoder runout
    SET_PIN PIN=Beep VALUE=1
        UPDATE_DELAYED_GCODE ID=turn_off_beep DURATION=8
insert_gcode:
    M117 Filament encoder inserted
event_delay: 3.0
pause_delay: 0.8
```

IdeaFormer

For Marlin 2.0/ Klipper firmware




IdeaFormer

IdeaFormer

